When do single-cell response distributions improve drug retrieval?

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Abstract

Artificial intelligence virtual cells aim to predict perturbation responses and identify drugs that drive cells towards a desired state. Most approaches rank drugs using average gene-expression responses, although single-cell measurements retain variation across cells. Whether this detail improves drug choice remains unclear. Here we introduce population-to-population drug retrieval (PopRetrieve) to test when response distributions improve ranking beyond average responses. Across perturbation datasets, magnitude-aware means recovered most gains over directional matching, leaving a smaller contribution from cell-to-cell structure. Independent functional associations were also largely recovered by magnitude-aware means. The same held-out protein measurements favoured different ranking strategies when evaluated as averages or full populations. Across disjoint cellular states, strongly correlated drug-response similarities nevertheless selected different closest response matches. With candidate responses observed, population retrieval gave a small gain beyond magnitude-aware means. With responses predicted in a new context, it showed no supported gain over directional matching across six predictors, although Gaussian optimal transport retained a gain over magnitude-aware means. These results show that the value of higher cellular resolution depends on the information retained by both retrieval and evaluation, and on whether prediction preserves the response features that change candidate ranking.

Introduction

Artificial intelligence virtual cells aim to predict how cells respond to genetic or chemical perturbations and, ultimately, to identify drugs that can drive cells towards a desired state[1–3]. A common strategy is to rank candidate drugs by comparing their gene-expression responses with a target response[4,5]. Most such approaches reduce each perturbation to an average expression signature, making large perturbation libraries tractable but discarding variation among individual cells. Single-cell perturbation atlases now retain this variation at scale[6], revealing that the same perturbation can elicit different responses across cellular states[7] or leave a small resistant subpopulation[8]. Whether this additional cellular detail leads to better drug choices, however, remains unclear.

Greater cellular resolution does not necessarily translate into better decisions. Two perturbations can have similar average responses yet differ substantially in how individual cells respond, so full single-cell distributions can contain information that any average representation loses. But an apparent advantage of using response populations need not arise from cell-to-cell structure itself. Conventional directional matching discards response magnitude, whereas many population-level comparisons retain it, allowing gains that can be explained by the average response alone. Moreover, whether the additional information is useful depends on how success is evaluated. Better agreement with a transcriptomic response distribution need not imply better recovery of biological mechanism or functional similarity, and an evaluator that retains population structure may reward different information from one based only on average responses. The relevant question is therefore whether that additional information changes candidate ordering under an evaluation that reflects the intended use.

This question becomes more important in prospective applications, where candidate responses in a new cellular context are not observed and must first be predicted[9,10]. Recent perturbation models increasingly aim to generalise across compounds, doses, combinations and cellular contexts[11–14]. Population-level ranking can benefit from cell-to-cell response structure only if the predictor preserves the perturbation-dependent variation that matters for drug choice. A model may reproduce heterogeneous cells without accurately capturing how different cellular states respond to the perturbation. Thus, the value of higher cellular resolution depends on whether additional response structure changes candidate ranking and can be recovered in a new context.

Here we introduce population-to-population drug retrieval (PopRetrieve) to determine when single-cell response distributions improve drug ranking beyond average-response representations. We separate information carried by response direction, response magnitude and cell-to-cell structure, and test whether the resulting gains persist under biological and functional evaluation and when candidate responses must be predicted. Across multiple perturbation datasets, a magnitude-aware mean recovers most of the apparent advantage of population-level retrieval, whereas the additional contribution of cell-to-cell structure is smaller and depends on how success is evaluated. We further find that strongly correlated drug-response similarities across cellular states can still select different closest response matches. Population structure gives a small retrieval gain beyond magnitude-aware means when candidate responses are observed. With predicted responses, the gain depends on both the predictor and the mean reference. Together, these comparisons separate retrieval gains associated with response magnitude, evaluator resolution and candidate-response prediction.

Results

Population retrieval retains response information beyond mean signatures

Population-to-population retrieval ranks candidates by comparing query and candidate response populations directly (Fig. 1a,b), whereas classical mean-signature retrieval[4,5] first collapses each population to a single mean differential-expression vector. Retaining the cells makes response structure available that averaging removes, including differences between cellular states or subpopulations. The ranking task is the same; the representation determines how much measured response reaches the decision. Two candidate populations can share the same mean response and still differ markedly in how individual cells respond (Fig. 1c,d). A population-aware score can separate such candidates because it retains distributional structure that any mean-only score discards. Whether that additional information matters for retrieval is a decision question: it matters only when it changes relative candidate scores enough to alter candidate ordering (Fig. 1e). These representations form an information ladder: directional matching

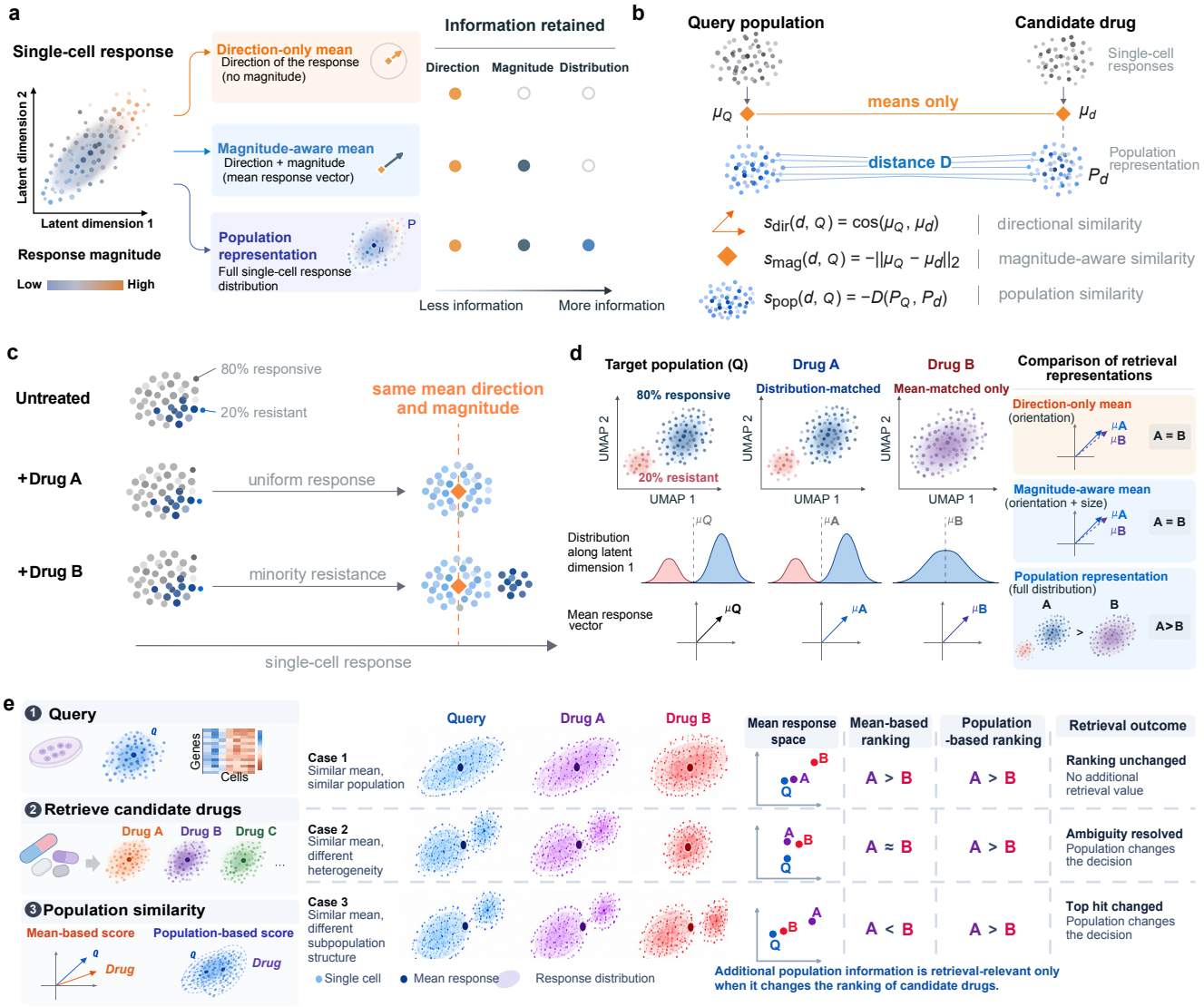


Figure 1: **Single-cell response representations retain different information and can change candidate ordering.** **a**, Three representations of the same single-cell perturbation response retain progressively more information: response direction only, direction and magnitude of the control-referenced mean response, and the full raw treated-cell response population. Filled and open circles indicate retained and discarded information. **b**, Corresponding retrieval rules compare query and candidate responses using directional similarity, magnitude-aware distance between control-referenced mean responses, or a distance between raw treated-cell response populations. **c**, Two perturbations can have the same mean response direction and magnitude while differing in their single-cell response structure. Drug A produces a relatively uniform response, whereas drug B leaves a resistant minority population. **d**, A target population and two candidates illustrate the resulting ambiguity. Drug A matches the target distribution, whereas drug B matches only its mean; the direction-only and magnitude-aware mean scores therefore tie, while the population representation prefers drug A. **e**, Population information affects retrieval only when the additional response structure changes candidate ordering. Across three illustrative cases, population-based scoring either preserves the mean-based top-ranked candidate or resolves ambiguities that are invisible to mean-based matching.

retains response orientation, magnitude-aware means additionally retain response strength, and population comparison can also retain cell-to-cell response structure (Fig. 1a,b). For energy distance, population comparison reduces to mean-L_2 ranking as within-population variation vanishes under a common reference (Methods). Subpopulation-coverage scores additionally test whether candidates match all query states (Supplementary Table 1).

A magnitude-aware mean recovers most gains over directional matching

Population-level retrieval substantially outperformed directional matching, but most of the gain did not require the full response population. Across seven controlled-mixture task–setting combinations with observed candidate responses, energy-distance retrieval reached a macro-averaged Hit@1 (the fraction

of queries with the correct candidate ranked first) of 0.836, compared with 0.388 for mean cosine and 0.788 for magnitude-aware mean L_2 (Fig. 2a). The additional distributional benefit varied across tasks. In Frangieh Perturb-CITE-seq[15], magnitude-aware mean L_2 performed best, and energy distance fell below both mean baselines (Fig. 2b). Under partial observation, population scores again improved on directional matching, but not on the magnitude-aware mean. Worst-case subpopulation coverage reduced response-matching regret by a median of $+0.0304$ relative to mean cosine (95% drug-cluster bootstrap CI $+0.0155$ to $+0.0495$; Fig. 2c). Regret measures the loss in response matching relative to the best candidate in the available library (Methods).

The energy-distance advantage over directional matching also varied with minority-state abundance and cell line (Fig. 2d). Across five population scores, all improved on mean cosine, whereas none showed a positive median gain over magnitude-aware mean L_2 under partial observation (Fig. 2e). Matched-control referencing left the observed-population comparisons essentially unchanged. Under partial observation it preserved the response-matching gain but changed the top-ranked candidate in 88.5% of queries for worst-case coverage (Supplementary Note 1). Mean-L_2 scores also correlated more strongly with population scores than with directional scores (Fig. 2f). A magnitude-aware mean score therefore recovers most of the gain beyond directional scoring, while the additional benefit of the full population is smaller and not consistently supported across evaluation settings. Because both criteria reward response correspondence, the next analysis tested whether the increment survives biological and functional evaluation.

Population-level gains depend on how success is evaluated

The value of that information depends on the evaluation criterion. The same rankings were evaluated by response matching and biological or functional criteria (Fig. 3). Response-matching gains did not consistently translate into recovery of drugs with the same mechanism of action. Across 600 leave-drug-out queries, worst-case subpopulation coverage improved response matching over mean cosine by a median of $+0.028$ (95% cluster-bootstrap CI $+0.0114$ to $+0.0510$). The same rankings gave a median mechanism-recovery gain of 0.000 and a mean of -0.024 (Fig. 3a,b). All five population scores showed this divergence between response matching and mechanism recovery (Supplementary Note 1). The mean-based minority-state proxy and the cell-level preservation measure selected different candidates (Fig. 3c). Response divergence did not reliably identify positive mechanism-recovery gains, and a pre-specified selection rule did not enrich for response-matching benefit (Fig. 3d–f; Supplementary Note 1).

Population-level retrieval also showed an association with the external functional readout. Using GDSC2[16,17] dose–response profiles from cell lines excluded from the transcriptional retrieval experiment, and removing the shared cell-line sensitivity effect, energy-distance retrieval reached a median per-query Spearman correlation of $+0.265$ with drug–drug functional similarity, against $+0.083$ for control-referenced mean cosine (Fig. 3g; Supplementary Table 3). Magnitude-aware mean L_2 reached $+0.285$; the median within-query difference between energy and mean L_2 was $+0.0012$ and varied in direction across the three cell lines. Response-magnitude matching alone reached $+0.232$ from a single scalar per perturbation. After adjusting for this magnitude channel, the remaining energy association was $+0.105$, positive in all three cell lines (Fig. 3g,h). Scored against absolute GDSC2 potency, energy and raw mean cosine were negative, whereas candidate magnitude alone was positive (Fig. 3i). Across queries, energy distance tracked candidate response magnitude at a median Spearman correlation of $+0.780$ (Fig. 3i). Thus, magnitude-aware means largely recovered the functional association of popula-

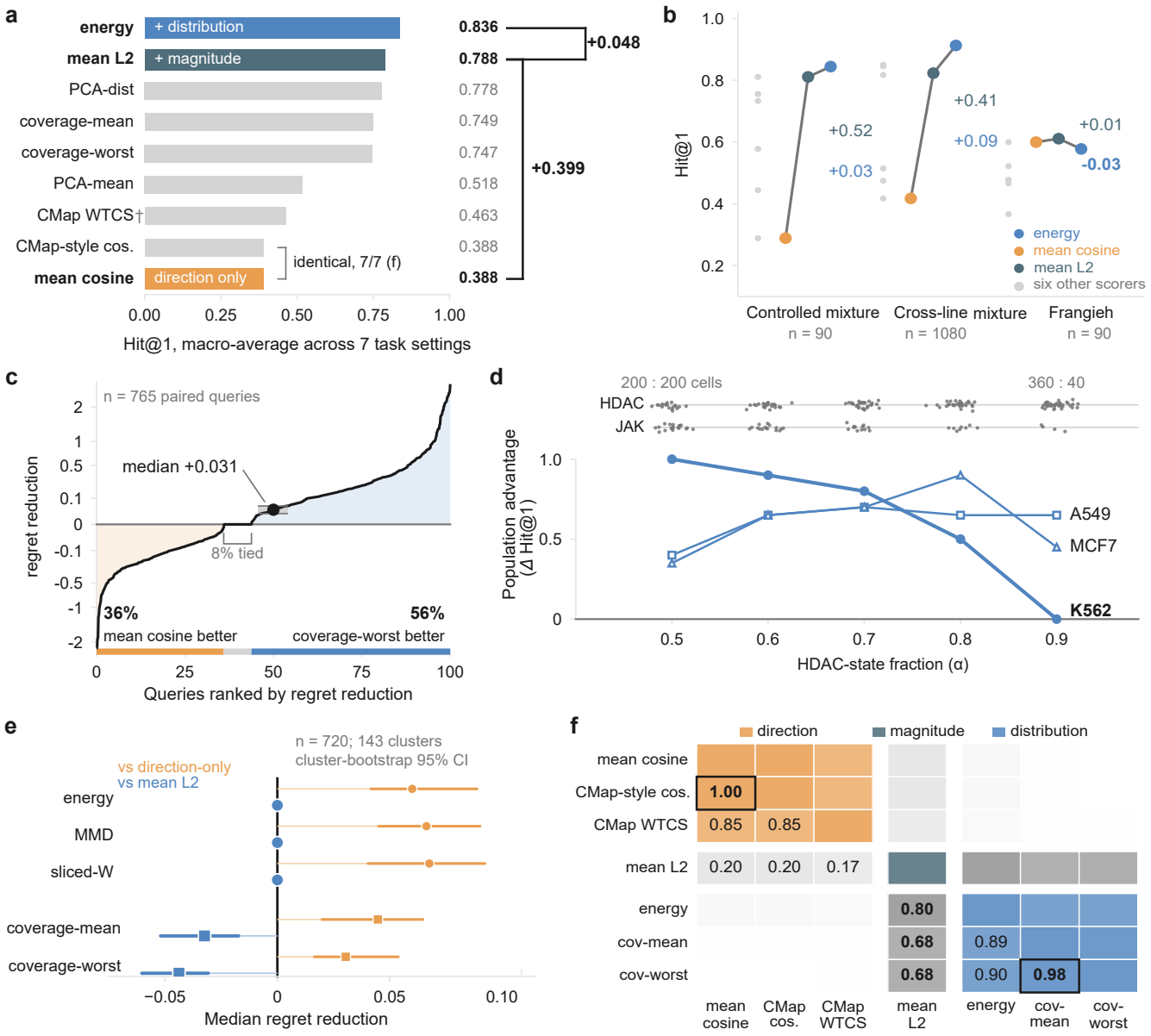


Figure 2: **A magnitude-aware mean score recovers most of the gain beyond directional scoring.** **a**, Hit@1 for nine scores, macro-averaged over seven task–setting combinations. Direction-only mean cosine, magnitude-aware mean L2 and energy distance reach 0.388, 0.788 and 0.836, respectively. Dagger, published CMap connectivity score (WTCS). **b**, The three scores by task ($n = 90$, 1,080 and 90 queries). The magnitude step is +0.52 and +0.41 for the two SciPlex3 tasks and +0.01 for Frangieh; the distributional step for Frangieh is −0.03. **c**, Worst-case subpopulation coverage versus direction-only mean cosine for 765 partial-observation queries. The curve shows the quantile function of per-query regret reduction, with positive values favouring coverage. Formal inference used 720 queries from 143 query-drug clusters and gave a median reduction of +0.0304 (95% cluster-bootstrap CI +0.0155 to + 0.0495). **d**, Energy advantage over mean cosine as a function of minority fraction α ($n = 20$ seeds per point). In K562 the advantage falls from +1.00 to 0.00; in A549 and MCF7 it ranges from +0.40 to +0.65 and from +0.35 to +0.45, respectively. **e**, Median regret reduction for five population-level scores across all 720 individual-drug queries from 143 query-drug clusters. All five scores improve on direction-only mean cosine (all 95% cluster-bootstrap CIs exclude zero). Relative to magnitude-aware mean L2, the three distributional scores have zero median regret reduction with CIs including zero, whereas the two coverage scores are negative (−0.030 and −0.041). **f**, Spearman correlations among seven scores for 54,180 query–candidate comparisons from 1,260 queries. Shade, $|\rho|$; rings, the closest pair within a tier. Mean L2 correlates more strongly with distributional scores ($\rho = 0.68$–0.80) than with direction-only scores ($\rho = 0.17$–0.20).

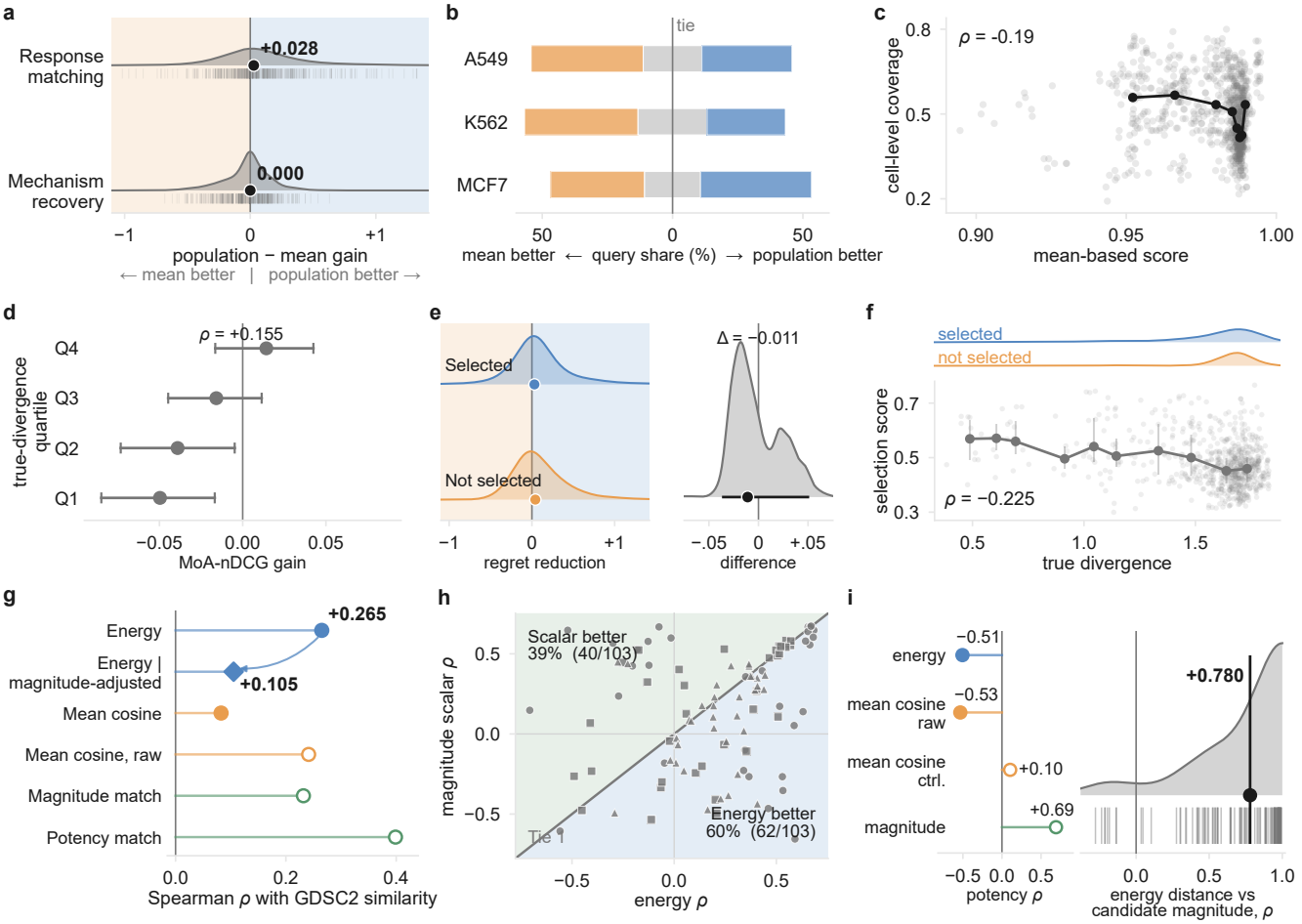


Figure 3: **The population advantage weakens, and can change direction, under biological and functional evaluation. a**, 600 leave-one-drug-out queries from 130 query-drug clusters, evaluated by response matching and mechanism recovery. Black points and thick lines show medians and 95% cluster-bootstrap CIs: +0.028 (+0.0114 to + 0.0510) and 0.000 (0.0000 to 0.0000), respectively. **b**, Directional composition of mechanism-recovery gain for 200 paired queries per cell line. Warm, grey and cool segments denote decreased, tied and increased recovery under population retrieval, respectively: 86/45/69 in A549, 87/53/60 in K562 and 72/43/85 in MCF7. Across one median effect per held-out drug ($n = 130$), two-sided Wilcoxon $P = 0.119$. **c**, Comparison of a mean-based minority-state similarity score with a cell-level nearest-neighbour measure in 860 candidate cases from 20 controlled-mixture seeds. Grey points are candidate cases; the black line joins medians in eight equal-count bins. Pooled Spearman $\rho = -0.188$; the two measures selected different top-ranked candidates in 18 of 20 seeds. **d**, Population-minus-mean MoA-nDCG@10 gain by response-divergence quartile. Points and intervals are quartile means and 95% query-drug-cluster bootstrap CIs; cluster-bootstrap Spearman $\rho = +0.1548$ (95% CI +0.0722 to +0.2349). **e**, Regret-reduction distributions for 584 queries selected by a pre-specified rule and 125 unselected queries; right, their cluster-bootstrap median difference across 141 query-drug clusters (−0.0109, 95% CI −0.0355 to +0.0498). **f**, Selection score versus measured response divergence among 720 individual-drug queries ($\rho = -0.2247$, 95% cluster-bootstrap CI −0.3144 to −0.1289); top, divergence distributions for selected and unselected queries. **g**, Median ranking correlations with GDSC2 drug–drug functional similarity ($n = 103$). Stems begin at zero; the arc links energy retrieval to its magnitude-adjusted value. **h**, Per-query comparison of energy distance with the magnitude-matching scalar; energy wins on 62 queries, the scalar wins on 40, and one query ties. **i**, Left, median ranking correlations with absolute GDSC2 potency: energy and raw mean cosine are negative (−0.506 and −0.533), whereas control-subtracted mean cosine and magnitude are positive (+0.105 and +0.692). Right, per-query Spearman correlation between energy distance and candidate magnitude; median +0.780.

tion retrieval. Its value depends on how success is measured, including criteria beyond transcriptomic response matching.

Evaluator resolution changes the preferred retrieval strategy

We tested this dependence using a biological measurement unavailable to the retrieval methods. Frangieh Perturb-CITE-seq[15] measures RNA and 20 surface proteins in the same 218,331 cells, allowing RNA-derived rankings to be judged using protein responses withheld from every retrieval method. To separate evaluator resolution from score geometry, we constructed parallel mean- and population-level evaluators from the same protein measurements. The mean evaluator used the negative Euclidean distance

between control-referenced mean protein responses, whereas the population evaluator used the negative energy distance between the corresponding protein-response populations. We then evaluated two fixed RNA-derived rankings from the same representation ladder: magnitude-aware mean-L2 retrieval and population-level energy retrieval (Fig. 4a).

Under the protein mean-L2 evaluator, magnitude-aware mean RNA retrieval performed better than population RNA retrieval, with median per-query Spearman correlations of $+0.4255$ and $+0.4150$, respectively; the median paired difference, energy minus mean-L2, was -0.0208. Under the protein population-energy evaluator, the ordering reversed: population RNA retrieval reached $+0.3930$ against $+0.3199$ for magnitude-aware mean retrieval, with a median paired difference of $+0.0538$ (Fig. 4b). The paired difference had the same sign in all three immune conditions: -0.0105 versus $+0.0466$ in Co-culture, -0.0363 versus $+0.1342$ in Control, and -0.0318 versus $+0.0384$ in IFNγ for the mean and population evaluators, respectively (Supplementary Table 5). Thus, the same held-out protein measurements favour different RNA-side retrieval strategies depending on whether the evaluator retains only the mean response or the full response population, even when score geometry is matched.

Response magnitude alone explained part, but not all, of this dependence. A scalar ranking that compares only the overall magnitude of the query and candidate RNA responses reached median correlations of $+0.3223$ under the protein mean-L2 evaluator and $+0.2764$ under the protein population-energy evaluator, compared with $+0.1206$ under the direction-only protein-cosine reference. These values remained below the corresponding matched retrieval strategy under each primary evaluator: $+0.4255$ for magnitude-aware mean RNA retrieval under the mean evaluator and $+0.3930$ for population RNA retrieval under the population evaluator (Fig. 4c). The evaluator-form reversal therefore cannot be reduced to response magnitude alone.

Correlated state-specific responses can change the closest response match

HIR-Bench provides a controlled example in which different population-level objectives can favour different candidates, with an analytical comparison between majority- and minority-specialist candidates (Methods; Fig. 4d). Drug-response identity was recoverable within naturally heterogeneous glioblastoma compartments[7]. Malignant and myeloid compartments were analysed separately against compartment-matched controls. A supervised probe recovered within-patient drug-response identity more accurately than the best unsupervised procedure, with a median paired gap of $+0.117$ (Fig. 4e; Supplementary Note 3).

Global response similarity and the closest response match capture different aspects of candidate ordering. Across 18 within-patient drug pairs from four tumours, myeloid- and malignant-response similarities were strongly correlated ($\rho = +0.835$; Fig. 4f). Nevertheless, in the one patient with six jointly evaluable drugs, the two compartments selected different closest response matches for all six queries against the other five drugs. The other three patients each supplied a single alternative per query. In Tahoe-100M, response identity was also recoverable (supervised ceiling 0.853; best unsupervised procedure 0.611), and drug-pair similarities were strongly correlated across cell-cycle and control-derived states (median $\rho = 0.841$ and 0.778). Yet the median within-cell-line agreement of the first-ranked response match was only 42.4% and 39.3%, respectively, over 3,620 queries in 44 lines and 3,722 queries in 45 lines (Supplementary Note 3). Strong global correlation therefore coexisted with frequent changes in the closest response analogue.

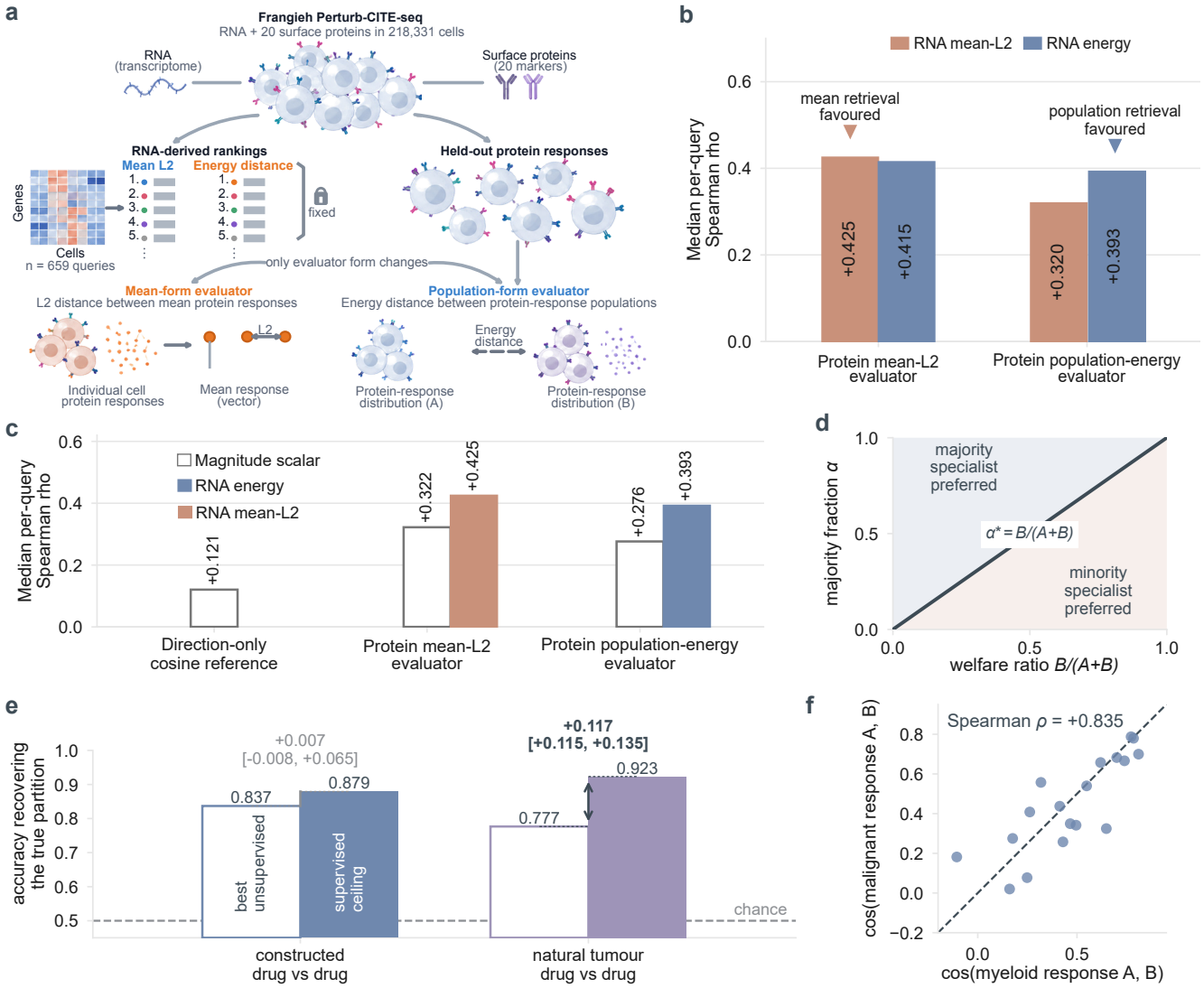


Figure 4: **Evaluator resolution changes retrieval preference, and disjoint compartments retain correlated response similarities. a–c**, The same fixed RNA-derived rankings evaluated using held-out protein responses from Frangieh Perturb-CITE-seq. **a**, RNA and 20 surface proteins are measured in the same 218,331 cells; protein measurements are unavailable to the retrieval methods. Two primary protein evaluators are constructed from the same cells: a magnitude-aware mean evaluator using negative Euclidean distance between control-referenced mean protein responses, and a population evaluator using negative energy distance between protein-response populations. **b**, Median per-query Spearman correlation across $n = 659$ queries for magnitude-aware mean-L2 and population-energy RNA retrieval. Under the protein mean-L2 evaluator, RNA mean-L2 and RNA energy reach $+0.4255$ and $+0.4150$, respectively, with a median paired difference of -0.0208 (energy minus mean-L2). Under the protein population-energy evaluator, the ordering reverses: $+0.3199$ and $+0.3930$, with a median paired difference of $+0.0538$. The paired difference has the same sign in Co-culture (-0.0105, $+0.0466$), Control (-0.0363, $+0.1342$) and IFNγ (-0.0318, $+0.0384$) for the mean and population evaluators, respectively. **c**, A scalar RNA response-magnitude ranking reaches $+0.3223$ and $+0.2764$ under the protein mean-L2 and population-energy evaluators, respectively; its value under the direction-only protein-cosine reference is $+0.1206$. **d**, HIR-Bench boundary $\alpha^* = B/(A+B)$ at which mixture-weighted mean welfare switches preference between majority- and minority-specialist candidates (Supplementary Note 2). **e**, **f**, Four glioblastoma tumours analysed in disjoint malignant and myeloid compartments with matched drugs and controls. **e**, Recovery of within-patient drug-response identity; open bars, best unsupervised method; filled bars, supervised probe. Median paired gap, $+0.117$ (95% patient-level cluster-bootstrap CI $+0.115$ to $+0.135$, $n = 36$ splits across four patients), versus $+0.007$ in the constructed comparator. **f**, Myeloid-response similarity versus malignant-response similarity across compartments (Spearman $\rho = +0.835$, $n = 18$ drug pairs); dashed line, equal similarity.

Observed candidate responses reveal a small population-level gain beyond the mean

Prospective drug retrieval ultimately requires ranking compounds whose effects in the target context have not been measured. We first established the information ceiling by making those candidate responses directly available, before replacing them with predictions in Fig. 6. In the Tahoe-100M intervention-identification task[18], untreated cells and a desired treated population define a query for ranking a fixed 92-compound library (Fig. 5a). We evaluated the true intervention's rank across 3,992 queries from 44 cell lines. The observed and predicted arms separate the value of population information from the cost of forecasting it. With candidate responses observed, mean reciprocal rank (MRR), which increases as the true intervention moves towards the top of the ranking, increased from 0.945 for directional mean cosine to 0.965 for magnitude-aware mean L_2 and 0.976 for energy-population retrieval (Fig. 5b). The additional population-level gain depended on the score: maximum mean discrepancy reproduced it, whereas sliced Wasserstein distance performed below both mean baselines (Supplementary Note 1). Energy-population retrieval also changed individual decisions. Relative to magnitude-aware mean L_2, it corrected the top-ranked drug on 2.0% of queries and broke a correct choice on 0.5% (Fig. 5c).

The oracle gain was concentrated in queries for which directional mean matching left room to improve (Fig. 5d). A ceiling-preserving permutation null nevertheless showed that headroom alone did not identify population-beneficial cases (Fig. 5e; Methods). Among measured query properties, response detectability was associated with population-route performance after conditioning on directional mean-cosine retrieval. Neither cross-fitted drug-by-state interaction nor its reproducibility showed a supported association (Fig. 5f–h; Methods). Population information therefore had measurable value with observed responses, but limited room to improve retrieval.

Predicted-response gains depend on the predictor and mean reference

Replacing observed candidate responses with predictions from the primary average-effect model reverses the advantage over directional mean cosine (Fig. 6a). MRR decreases by 0.058 for direction-only mean cosine and by 0.124 for energy-population retrieval (Fig. 6b). The MRR difference between energy-population retrieval and direction-only mean cosine is then -0.0356 (95% CI -0.0620 to -0.0115; Fig. 6c). None of the six forward predictors shows a supported energy-retrieval advantage over directional mean cosine. The Gaussian optimal-transport map has a positive point estimate against that reference ($+0.0027$, 95% CI -0.0215 to $+0.0243$), and retains an advantage over magnitude-aware mean L_2 ($+0.0110$, $+0.0087$ to $+0.0134$; Supplementary Table 6).

Mean-response magnitude accounts for most of the average-effect predictor's retrieval deficit. On the same predictions, magnitude-aware mean L_2 has an MRR difference of -0.0335 relative to direction-only mean cosine, while energy-population retrieval has a difference of -0.0020 relative to magnitude-aware mean L_2 (-0.0036 to -0.0005; Fig. 6d). The additive predictor translates the query line's vehicle cells by an effect averaged over the other 43 lines, preserving baseline heterogeneity and generating zero drug-by-state interaction (Fig. 6e; Supplementary Note 4).

A state-conditioned average-effect predictor generates state-specific interaction while preserving direction-only mean-cosine MRR (0.8883 against 0.8876 for the average-effect predictor). Energy-population retrieval has an MRR difference of -0.0314 relative to direction-only mean cosine (-0.0584 to -0.0076; Fig. 6f). Of the 755 queries on which observed-response energy retrieval gained over mean cosine, the average-effect predictor retains a gain on 209 and loses it on 546 (Fig. 6g). Observed

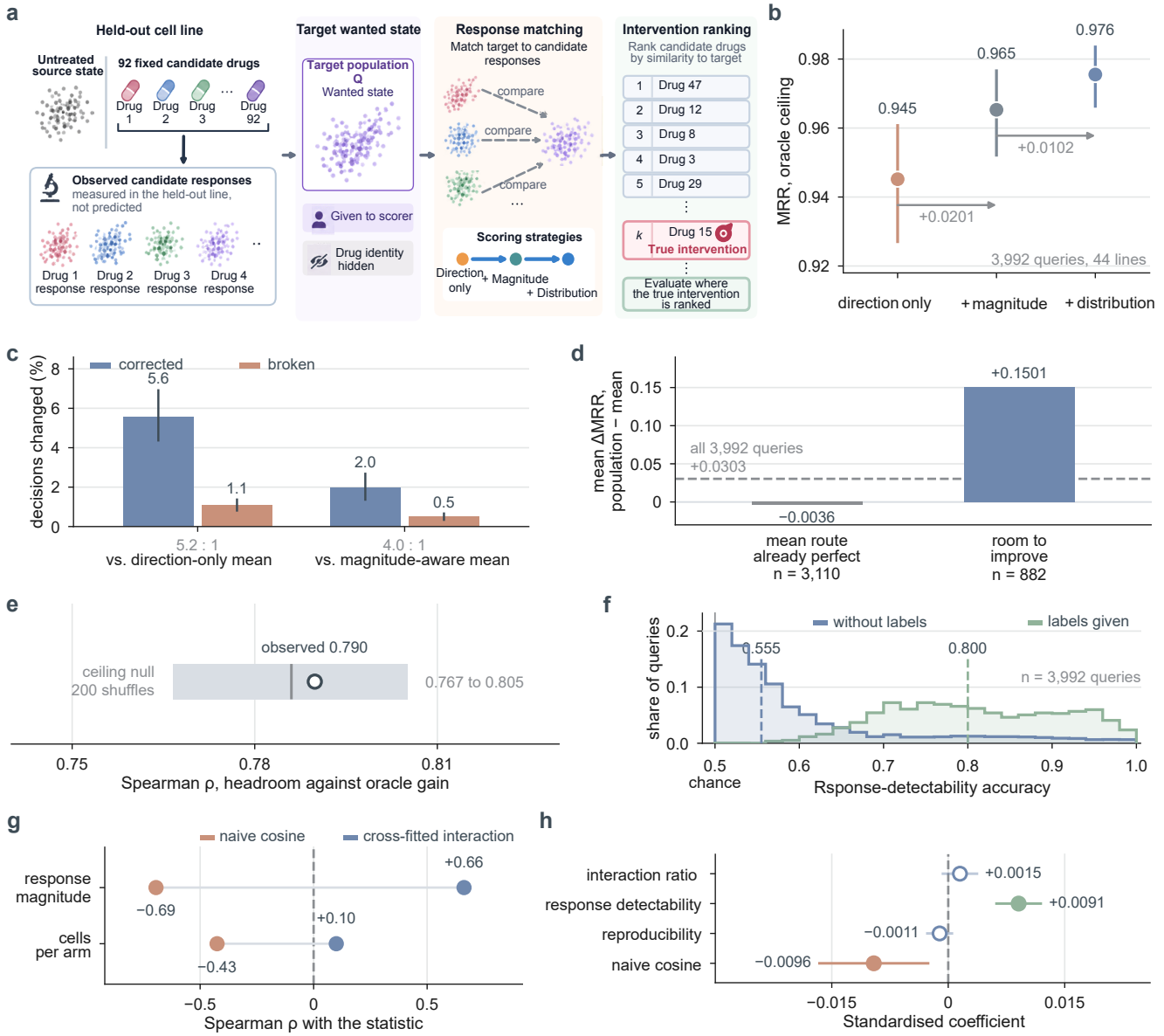


Figure 5: **Population structure adds a small oracle retrieval gain beyond response magnitude.** Source-to-target intervention identification on Tahoe-100M: 3,992 queries over 44 cell lines and a fixed 92-compound library, with candidate responses observed within the query line; predicted candidates are evaluated in Fig. 6. **a**, A held-out line's untreated cells and a desired treated population define the query; candidates comprise all 92 compounds. **b**, Mean reciprocal rank for the three scoring rules in Fig. 1a, with 95% bootstrap intervals: 0.945, 0.965 and 0.976. The two increments are +0.0201 and +0.0102. **c**, Top-1 decisions changed relative to each mean reference; ratios show the relative frequency of the two directions. **d**, The pooled +0.0303 advantage stratified by direction-only mean-cosine headroom; on 3,110 of 3,992 queries it already ranks the true drug first at every seed. **e**, Spearman correlation between direction-only mean-cosine headroom and population gain, with a within-context permutation null. **f**, Single-cell response detectability for a query drug versus vehicle cells from the same line, using up to 300 cells per arm; median accuracy 0.800 with labels and 0.555 without labels. **g**, Interaction statistics versus response magnitude and cells per arm. The naive $1 - \cos(r_1, r_2)$ gives −0.69 and −0.43; cross-fitted S_{int} gives +0.66 and +0.10. **h**, Standardised coefficients from regressions of energy-population reciprocal rank, with direction-only mean-cosine reciprocal rank as a covariate; the first three rows use $n = 3{,}618$ queries in 44 cell-line clusters and the naive cosine uses $n = 3{,}620$. Filled markers, intervals excluding zero.

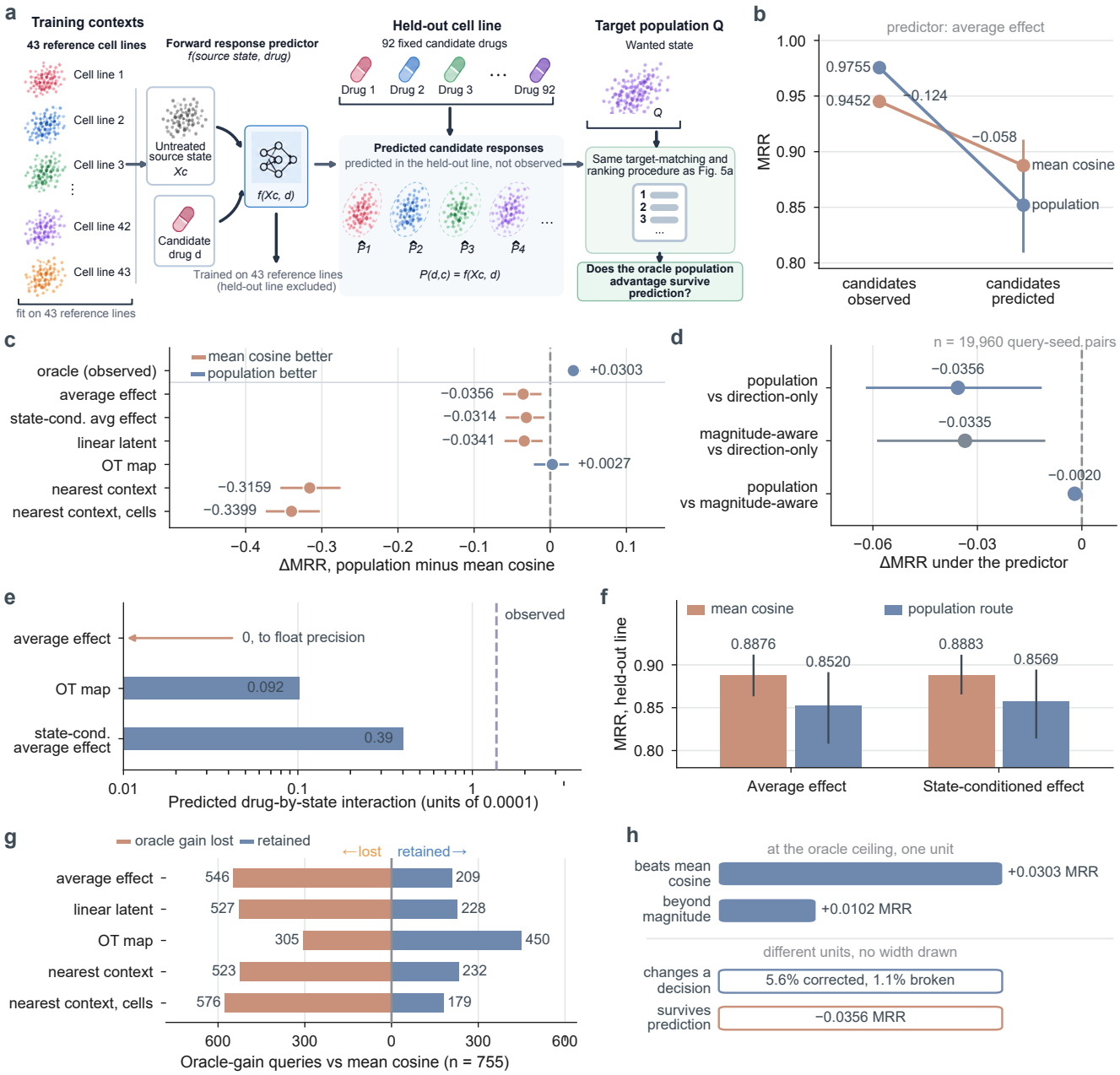


Figure 6: **Predicted-response retrieval gains depend on the mean reference.** The task, queries, library and split are as in Fig. 5; candidate responses in the held-out line are produced by a forward predictor fitted on the other 43 lines. **a**, The intervention-identification task with predicted candidate responses. **b**, MRR with observed and predicted candidates under the primary average-effect predictor; direction-only mean cosine decreases by 0.058 and energy-population retrieval by 0.124. **c**, ΔMRR of energy-population retrieval over direction-only mean cosine, with observed candidates and for six predictors; the difference changes sign under the average-effect predictor. **d**, The same predictions judged three ways: energy-population retrieval's −0.0356 deficit against direction-only mean cosine is matched by the magnitude-aware mean's −0.0335, and −0.0020 remains once magnitude is controlled. **e**, Median drug-by-state interaction from each predictor versus the cross-fitted value measured in real cells (dashed); the additive predictor generates zero interaction. **f**, A state-conditioned average-effect predictor generates interaction while preserving direction-only mean-cosine accuracy, but energy-population retrieval remains lower. **g**, Among the 755 queries on which oracle energy-population retrieval gained over direction-only mean cosine, 209 retain a gain under the primary average-effect predictor and 546 do not. Additional gains occur among queries with no oracle gain: 343 for the primary predictor and 1,165 for the Gaussian OT map. **h**, MRR budget across Figs. 5 and 6; the first two rows show oracle gains over mean cosine and mean L2, respectively. The final rows show the fraction of oracle mean-cosine gains retained and the primary average-effect predictor's signed gain over mean cosine.

candidate responses therefore support a small population-level gain beyond magnitude-aware means, whereas predicted-response gains depend on the predictor and mean reference (Fig. 6h).

Discussion

Our results separate two ideas that are often treated as equivalent: preserving more cellular detail and making a better intervention decision. Single-cell response populations retain variation that is necessarily lost by averaging, but only a fraction of this information changes which drug is preferred. Across the settings examined here, much of the apparent advantage of population-level retrieval over directional matching was already recovered by a magnitude-aware mean representation, leaving a smaller and less consistent contribution from cell-to-cell response structure. This distinction became clearer as evaluation moved beyond transcriptomic response matching. Population-level gains did not consistently carry over to mechanism recovery or independent functional measurements, and the same held-out protein measurements favoured different retrieval strategies depending on whether they were evaluated as average responses or as full single-cell populations. Strongly correlated drug-response similarities across cellular states nevertheless selected different closest response matches. Together, these results show that greater cellular resolution does not automatically translate into greater decision value. What matters is whether the additional information changes candidate ordering under an evaluation that reflects the intended biological use.

This distinction becomes more consequential when single-cell responses must be predicted rather than measured. In prospective use, a virtual-cell model must infer how each candidate drug would alter a cellular population that has not yet been experimentally perturbed. Our observed-response experiments show that population structure can contain information beyond a magnitude-aware mean that is sufficient to change drug ranking. The prediction experiments, however, show that preserving cellular heterogeneity is not equivalent to preserving this decision-relevant information. Under the primary predictor, most of the loss in population-level retrieval was already present at the magnitude-aware mean level, indicating that the bottleneck was not exclusively distribution-specific. Explicitly introducing state-dependent perturbation effects did not restore the gain over directional mean cosine, whereas Gaussian optimal transport retained a gain over magnitude-aware mean L_2. Models can therefore reproduce heterogeneous cells, or even measurable state dependence, without preserving the particular perturbation-dependent structure that changes candidate ordering. The relevant predictive target is the response structure that alters intervention choice. None of the predictors examined here simultaneously achieved strong generalisation to the target context and faithful recovery of the state-specific perturbation effects observed in real cells, leaving open whether more capable conditional models can preserve the population-level gain seen with observed responses.

More broadly, these results suggest that increasing biological resolution should be treated as a hypothesis about downstream decision value rather than as an objective in itself. For intervention selection, a higher-resolution representation provides additional value only when the added information is reproducibly supported by the measurement, cannot be recovered from a simpler representation, changes candidate ordering and, when prediction is required, can be preserved in a new context. These requirements separate four questions that are often conflated in virtual-cell evaluation: what biology is measurable, what information a representation retains, what information changes a decision and what information a model can predict in a new context. These results also suggest that evaluator resolution should be treated as part of the task definition rather than as a neutral implementation choice. The relative importance of these requirements will depend on the perturbation library, cellular context, measurement modality and

downstream objective, and the external criteria examined here capture biological or functional similarity rather than therapeutic benefit. The goal of higher-resolution virtual-cell modelling is to identify and predict cellular variation that is biologically supported and consequential for the decision the model is intended to inform.

Methods

Study design and data resources

The primary observed-population retrieval analyses used SciPlex3[6], comprising 276,325 cells across three cancer cell lines (A549, K562 and MCF7), 188 compounds shared across the three contexts and a 2,000-gene expression space. SciPlex3 supports both within-context retrieval and controlled cross-context constructions because the same compound library is measured across all three cell lines. Frangieh melanoma Perturb-CITE-seq[15] spans Control, IFNγ and Co-culture conditions with 239 CRISPR knockouts. Both analyses used the same RNA matrix with different inclusion criteria: the retrieval analysis uses the 218,023 cells left after dropping nine knockout targets with fewer than 50 cells, and the evaluator-shape experiment uses all 218,331 cells of the RNA–protein barcode intersection, of which 215,915 contribute to at least one condition mean once the 60-cell floor per knockout is applied. Frangieh provides an independent perturbation-retrieval setting and matched surface-protein measurements for the evaluator-shape experiment, in which the protein modality is withheld from every retrieval method and used for evaluation.

Additional resources serve roles outside the primary retrieval benchmark. ZhaoSims2021[7] contains drug-treated patient-derived glioblastoma tissue and supplies natural-tissue diagnostics of state-dependent responses and their recoverability. Tahoe-100M[18] provides larger-scale measurements of the same information requirements under intrinsic within-cell-line heterogeneity and is the substrate for the source-to-target intervention-identification benchmark of Figs. 5 and 6. We analyse plate 3, which comprises 4,158,278 cells across 50 cell lines and 92 compounds plus vehicle, using the released 2,304-gene log1p-transformed representation. Finally, GDSC2 dose–response profiles provide an external functional readout for drug–drug similarity and are joined to the SciPlex3 compound panel independently of the transcriptomic retrieval scores. Dataset scale, inclusion criteria and analysis roles are summarised in Supplementary Table 2.

Preprocessing and construction of perturbation-response populations

Single-cell expression analyses were performed in log-transformed expression space using the processed gene sets defined for each dataset. SciPlex3 and Frangieh were analysed in 2,000-gene representations. For SciPlex3, cells were first capped at 120 per compound, cell line and dose and at 3,000 controls per cell line, using seed 0. From those retained cells, 60,000 cells were sampled without replacement with the same seed; raw counts were normalised to a total of 10^4 per cell, log1p-transformed, and 2,000 highly variable genes were selected using the Seurat highly-variable-gene criterion[19] as implemented in Scanpy[20]. The final SciPlex3 matrix was then built from raw counts on those genes, normalised with all-gene size factors. For Frangieh, no HVG subsample was used: all cells passing the immune-condition and perturbation filters were normalised to 10^4 counts per cell, log1p-transformed, and reduced to 2,000 genes by the same Seurat criterion. For ZhaoSims2021, counts were normalised to a total of 10^4 per cell, log1p-transformed and reduced to 2,000 highly variable genes. Tahoe-100M was used in the released preprocessed form (2,304 highly variable genes, log1p transformed), with no additional feature selection or representation fitting.

For a perturbation d measured in cellular context c, let

$$X_{d,c} = \{x_1^{(d,c)}, \ldots, x_{n_{d,c}}^{(d,c)}\} \tag{1}$$

denote its treated single-cell expression profiles and let

$$\mu_{0,c} = \frac{1}{n_{0,c}} \sum_{j=1}^{n_{0,c}} x_j^{(0,c)} \tag{2}$$

denote the mean of the matched control population. The conventional perturbation signature is the control-referenced mean response

$$\bar{d}_{d,c} = \frac{1}{n_{d,c}} \sum_{i=1}^{n_{d,c}} x_i^{(d,c)} - \mu_{0,c}. \tag{3}$$

Mean-signature retrieval uses the control-referenced mean response, whereas population-level retrieval scores receive the raw normalised-expression treated-cell matrices,

$$P_{d,c} = \left\{ x_i^{(d,c)} \right\}_{i=1}^{n_{d,c}}, \tag{4}$$

while variation among cells remains available to the population score. Mean cosine and mean L_2 each subtract the corresponding control mean from the treated-cell mean; query and candidate controls may therefore differ. Energy distance, RBF-MMD, sliced Wasserstein and coverage use the raw normalised-expression populations and do not subtract matched controls in the primary population comparisons. Matched-control-referenced population scores were evaluated separately as a sensitivity analysis (Supplementary Note 1). State- and compartment-specific responses use matched controls from the same biological context.

Population-to-population retrieval and signature baselines

Population-level retrieval. For query Q and candidate populations $\{P_d\}_{d=1}^{D}$, retrieval ranks the library by similarity or distance. Population scores use the empirical treated-cell populations; signature scores use their control-referenced mean responses. Each comparison uses the same queries and candidate library. We evaluated energy distance[21] (the primary population score), maximum mean discrepancy[22], sliced Wasserstein distance[23], and the mean and worst-case limits of a subpopulation-coverage score. All rank candidates from smallest to largest distance. For a candidate population $P = \{x_1, \ldots, x_m\}$ and a query population $Q = \{y_1, \ldots, y_n\}$ the energy distance is estimated as

$$\widehat{\mathcal{E}}(P, Q) = \frac{2}{mn} \sum_{i,j} \|x_i - y_j\| - \frac{1}{m(m-1)} \sum_{i \neq i'} \|x_i - x_{i'}\| - \frac{1}{n(n-1)} \sum_{j \neq j'} \|y_j - y_{j'}\|, \tag{5}$$

using the Euclidean norm. Energy and RBF-MMD use unbiased U-statistics, including within the coverage score. The secondary PCA-latent energy score in Fig. 2a uses the V-statistic, with within-population denominators m^2 and n^2. Sampling caps and secondary-score settings are in Supplementary Methods.

Subpopulation-coverage score. The query is partitioned into $K = 2$ components. Construction labels are used when available; otherwise query cells are partitioned by k-means and candidate cells assigned to query components as specified in Supplementary Methods. Writing d_k for the base distance in com-

ponent k, the coverage distance and ranking score are

$$D_\beta = \frac{1}{\beta}\Big(\log \sum_{k=1}^{K} e^{\beta d_k} - \log K\Big), \qquad S_\beta = -D_\beta. \tag{6}$$

The base distance is energy, MMD or sliced Wasserstein. As $\beta \to 0$, D_β averages component distances; as $\beta \to \infty$, it approaches the worst-matched component. Coverage-mean and coverage-worst use these limits; for $K = 1$, coverage equals the global distance.

Mean-signature baselines. Mean-signature retrieval ranks the control-referenced vectors $\bar{d}_{d,c}$ by cosine similarity or Euclidean distance. Mean L_2 differs from mean cosine in both magnitude sensitivity and score geometry. The implemented CMap-style cosine is numerically identical to mean cosine on these vectors. We also evaluated the rank-based L1000 weighted connectivity score (WTCS)[5] as a distinct signature baseline. For the PCA comparison, one 30-component basis was fitted per query to a uniform sample of up to 4,000 cells pooled from the query and all 43 candidates. This transductive fit used expression values without candidate-identity, ground-truth or subpopulation labels; query cells were disjoint from the correct candidate's cells. Control cells were excluded from fitting, and control means were projected through the fitted basis. Retrieval then compared latent-space means or populations.

Connection between mean and population retrieval. Under a shared or common reference, define the control-referenced response population $R_{d,c} = P_{d,c} - \mu_{0,c}$, and write it as

$$R = \mu + \lambda\epsilon, \qquad E[\epsilon] = 0, \tag{7}$$

where $\mu = \bar{d}_{d,c}$ is the control-referenced mean response, ϵ represents within-population variation and λ controls its magnitude. As $\lambda \to 0$, the response population collapses to a point mass at μ, and population comparison therefore loses all information beyond the mean. For two point masses, energy distance is $2\|\mu_Q - \mu_d\|_2$, giving the same ranking as mean L_2 under a common control reference. Query and candidate controls can differ in the primary comparisons, so the equivalence is exact only under a common control reference. Cosine additionally removes response magnitude. These limits describe the information retained by each representation; WTCS and the population distances remain distinct scoring functions[24].

HIR-Bench construction

HIR-Bench is a controlled two-subpopulation latent-welfare construction used to establish when within-population structure can change a preferred candidate. Each synthetic instance assigns welfare weights α and $1-\alpha$ to majority and minority subpopulations; query cell counts follow these weights with rounding and a five-cell minimum per state (Supplementary Note 2). Candidate response centres range from balanced to majority- or minority-specialist, with a conflict parameter controlling the extent to which the two subpopulations prefer different candidates. Per-subpopulation utility is negative expected squared distance to that subpopulation's latent target. Let d_M be the candidate preferred by the majority and d_C the candidate preferred by the minority, with

$$A = u(d_M, 0) - u(d_C, 0), \qquad B = u(d_C, 1) - u(d_M, 1). \tag{8}$$

The mean-welfare preference changes at

$$\alpha^* = \frac{B}{A+B}. \tag{9}$$

When $A + B > 0$, mean welfare prefers d_M above the boundary and d_C below it. The worst-subpopulation optimum instead maximises $\min_k u(d,k)$ over the full library (Supplementary Note 2).

Response-matching and partial-observation evaluation

Observed-population retrieval. When complete candidate response populations were available, retrieval was evaluated over heterogeneous queries. Each query is a population built by combining two response modes at mixing weight $\alpha \in \{0.5, 0.7, 0.9\}$, and the correct answer is the candidate that reproduces both modes at the same weight, built from the held-out half of the same cells so that no cell is shared between a query and its answer. Two further candidates carry one mode each, and 40 distractor populations complete a 43-candidate library.

The three tasks differ in what the two modes are. In the controlled SciPlex3 task they are the HDAC- and JAK-associated response sources within a single cell line at 10 μM, each pooling every compound annotated to that class, so the two single-mode candidates are HDAC-only and JAK-only populations and each distractor is an individual compound of another class. In the cross-line and Frangieh tasks they are one perturbation observed in two biological contexts, two cell lines or Control and IFNγ, so the two single-mode candidates are that perturbation observed in one context only and each distractor is a different perturbation mixed across the same two contexts at the same weight; cross-line perturbations pool all measured doses. The two single-mode candidates are each referenced to their own single-context control, whereas the correct answer and every distractor are referenced to the α-mixed control. The cross-line and Frangieh tasks use the most divergent perturbations passing a split-half reliability floor in both contexts and a cross-context response cosine below 0.9: 12 drugs per cell-line pair and 6 knockouts. Their retrieval estimates therefore describe a panel selected for cross-context divergence.

Populations contain 400 cells in the controlled task and 200 in the other two, sampled without replacement where possible and with replacement otherwise. Queries within a task setting share one fixed cell split and vary by perturbation or mechanism, α and resampling seed. The controlled and cross-line tasks use ten seeds and Frangieh uses five; queries within a setting are repeated observations. The both-mode candidate's rank is one plus the number of candidates with strictly higher scores. This rank defines Hit@1, Hit@5, MRR and nDCG@10. A single-mode candidate ranked first counts as a miss, including when it carries the correct perturbation in one context. Hit@1 therefore evaluates population composition as well as perturbation identity. The primary summary is an unweighted macro-mean over seven task–setting combinations. Query-weighted summaries and tie handling are given in Supplementary Methods.

Partial-observation protocol. Each score ranks the same partially observed candidate library, and the resulting ranking is evaluated under response matching and mechanism recovery. Candidates are ordered by a label-blind stable sort; Hit@1 uses the best rank attained by a member of the query's mechanism or target set. Three splits construct the partial observation, all on SciPlex3, and in each the hidden material never enters the candidate library[25]. In *leave-drug-out*, one drug is hidden, the query is that drug's own treated population (up to 200 cells) and the library is every other drug in the context. In *leave-MoA-out*, an entire annotated mechanism class is removed from the library and the query is one drug drawn from the removed class, so recovery requires a same-target surrogate among the remaining drugs. In *partial-library*, a fraction of the drugs is treated as measured, the query is built from the pooled

responses of the unmeasured drugs, and the retriever selects the next batch of candidates. In all three, query states are obtained by k-means with $k = 2$ on the query cells and the minority state is the smaller cluster; candidate populations are capped at 120–150 cells per drug.

The response-matching criterion is an energy-based welfare regret. Each candidate's welfare is the aggregate over query states s of its negative energy distance to that state, $U(d) = \mathrm{agg}_s\left(-\mathcal{E}(P_d, Q_s)\right)$ with the aggregator taken as the minimum in the worst-case form, and the regret of the selected candidate $\hat{d}$ is

$$\mathrm{Regret}(\hat{d}) = \max_{d \in \mathcal{L}} U(d) - U(\hat{d}) \;\geq\; 0, \tag{10}$$

where $\mathcal{L}$ is the observed library. For population scorer p and mean reference r, regret reduction is $\mathrm{Regret}(\hat{d}_r) - \mathrm{Regret}(\hat{d}_p)$; positive values indicate improved response matching. Figures 2c and 3a compare worst-case coverage with mean cosine; Fig. 2e compares every population scorer with mean cosine and mean L_2. The energy-based criterion evaluates response fidelity using information also available to the population scorer.

Task-proximal biological evaluation and validation of the state-similarity measure. Mechanism-of-action (MoA) recovery was measured by normalised discounted cumulative gain at rank 10 (nDCG@10)[26]. Binary relevance is one for candidates sharing the query's annotated mechanism and zero otherwise. DCG is $\sum_{r=1}^{\min(10,N)} y_r / \log_2(r+1)$, where N is library size; the ideal DCG places all available relevant candidates first, and nDCG@10 is their ratio. Evaluation used the 600 leave-drug-out queries; the 120 leave-MoA-out and 45 partial-library queries were excluded because their libraries contain no member of the query mechanism class. Among the 600, 21 queries lacked a same-mechanism candidate and received nDCG $= 0$ for every method. They were retained as ties; the exclusion sensitivity is reported in Supplementary Methods.

We assessed a mean-based minority-state similarity score defined as $\frac{1}{2}\{1 + \cos(\bar{P}_d, \bar{Q}_{\min})\}$, where $\bar{P}_d$ is the candidate mean response and $\bar{Q}_{\min}$ is the mean response of the less represented query state. The cell-level reference was the fraction of minority-query cells whose Euclidean distance to their nearest candidate cell did not exceed the median distance pooled over all candidates in that seed. The comparison used a K562 HDAC–JAK controlled mixture ($\alpha = 0.7$), 20 fixed seeds and 43 candidates per seed (860 candidate cases). We compared the two metrics by pooled Spearman correlation and by agreement of the top-ranked candidate. The mean-based score did not agree with the cell-level measure and was not used as a biological endpoint.

External functional evaluation using GDSC2

We evaluated SciPlex3 rankings against drug–drug functional similarity derived from GDSC2 dose–response profiles[16,17] (release 8.5; Data availability). SciPlex3 compounds were matched to GDSC2 by normalised drug name, with substring matching used only as a fallback. After removal of one duplicate mapping and restriction to the GDSC2 screening platform, 35 compounds remained. For each matched compound, its functional profile was the vector of fitted dose–response AUC values across GDSC2 cell lines. The three cell lines used for the transcriptomic retrieval experiments, A549, K562 and MCF7, were excluded from these profiles, leaving 966 cell lines.

Cell lines differ systematically in their overall sensitivity to drug treatment, which can induce positive correlations between otherwise unrelated compound profiles. We therefore removed this shared cell-line

effect before calculating drug–drug similarity. For cell line l and compound d, the centred response was

$$\widetilde{\mathrm{AUC}}_{d,l} = \mathrm{AUC}_{d,l} - \frac{1}{|\mathcal{D}_l|} \sum_{d' \in \mathcal{D}_l} \mathrm{AUC}_{d',l}, \tag{11}$$

where $\mathcal{D}_l$ contains the full set of GDSC2 compounds measured in cell line l, including compounds outside the 35-drug SciPlex3 overlap. Functional similarity between query drug q and candidate drug c was then defined as

$$F(q,c) = \rho_S \left(\widetilde{\mathrm{AUC}}_{q,\bullet}, \widetilde{\mathrm{AUC}}_{c,\bullet} \right), \tag{12}$$

where ρ_S denotes Spearman correlation across cell lines shared by the two compounds. Drug pairs measured in fewer than 100 common cell lines were excluded. This yielded 591 usable drug pairs, with a median of 909 shared cell lines per pair. The corresponding uncentred functional similarity was retained as a sensitivity analysis.

For each SciPlex3 cell line, retrieval was evaluated in a leave-one-drug-out design. The 10 μM response population of the held-out compound served as the query, and the remaining matched compounds formed the candidate library, giving 30–34 evaluable candidates per query after valid-pair filtering. Each RNA-based retrieval rule produced a candidate ranking without access to GDSC2 data. Agreement between that ranking and the external functional ordering $F(q,c)$ was quantified by Spearman correlation across candidates. The RNA-derived rankings included energy-distance population retrieval, control-referenced mean cosine and magnitude-aware mean L_2, scored as $-\|\bar{d}_q - \bar{d}_c\|_2$. Within each cell line, query and candidate use the same control mean, so the mean-L_2 ranking is identical with or without control subtraction. Energy scoring used the U-statistic with at most 500 cells per drug and seed 42. To test whether apparent functional agreement required cell-to-cell distributional information, we additionally used a response-magnitude matching score,

$$M(q,c) = -\left| \|\bar{d}_q\| - \|\bar{d}_c\| \right|, \tag{13}$$

which prefers candidates whose overall transcriptional response magnitude is similar to that of the query. A query-independent magnitude score $\|\bar{d}_c\|$ was retained as a diagnostic control. An uncorrected mean-cosine score based on treated means without matched-control subtraction and a potency-matching diagnostic were included as sensitivity controls.

To estimate the component of the energy-based association not explained by response magnitude, we computed a partial rank correlation with the magnitude-matching channel removed. Retrieval scores and functional similarities were rank-transformed, each was residualised with respect to $M(q,c)$, and the correlation between the two residuals was reported as the magnitude-adjusted association. Overall values are median per-query Spearman correlations across 103 queries, with cell-line-specific medians as higher-level summaries. Queries share candidate pools. Energy-minus-mean-L_2 differences were calculated within each query and summarised overall and separately by cell line. Energy retrieval and the response-magnitude scalar are compared by per-query win counts (Fig. 3h).

Protein-based evaluator-shape experiment

Frangieh Perturb-CITE-seq[15] provides matched RNA and surface-protein measurements from the same cells, allowing the statistical form of an external evaluator to be changed while holding its biological measurement fixed. RNA and protein matrices were joined by cell barcode before analysis. Of the 24 measured antibodies, four isotype controls were removed after normalisation, leaving 20 surface-

protein features. Antibody-derived tag counts[27] were centred-log-ratio normalised within each cell, $y_{ij} = \ln(1 + x_{ij}) - m^{-1} \sum_k \ln(1 + x_{ik})$ for cell i and antibody j, with a pseudocount of one, a natural logarithm and the per-cell mean taken over all $m = 24$ measured antibodies including zero counts and the four isotype controls. All analyses were performed separately within the Control, IFNγ and Co-culture conditions to avoid mixing immune context with perturbation identity. Knockouts represented by at least 60 barcode-matched cells were retained, and populations were subsampled to at most 250 cells using random seed 42.

Protein-based evaluators. For knockout k in immune condition c, let $\bar{d}^{\text{prot}}_{k,c}$ denote its mean control-referenced protein response and let $D^{\text{prot}}_{k,c}$ denote the corresponding set of per-cell control-referenced protein responses. To separate evaluator resolution from score geometry, we defined three protein-side evaluators paralleling the RNA-side retrieval representations. The direction-only evaluator was

$$E_{\text{dir}}(q, d) = \cos\left(\bar{d}^{\text{prot}}_{q,c}, \bar{d}^{\text{prot}}_{d,c}\right). \tag{14}$$

The magnitude-aware mean evaluator was

$$E_{\text{mag}}(q, d) = -\left\|\bar{d}^{\text{prot}}_{q,c} - \bar{d}^{\text{prot}}_{d,c}\right\|_2. \tag{15}$$

The population evaluator was

$$E_{\text{pop}}(q, d) = -\mathcal{E}\left(D^{\text{prot}}_{q,c}, D^{\text{prot}}_{d,c}\right), \tag{16}$$

where $\mathcal{E}$ denotes the energy distance. Larger values therefore indicate a closer match for all three evaluators. The primary evaluator-resolution comparison contrasted E_{mag} and E_{pop}. These share the same Euclidean-distance geometry in the zero-variance limit, in which the energy distance between two point masses is proportional to the Euclidean distance between their means; E_{dir} was retained as a direction-only reference.

For all three definitions, protein responses were referenced to the matched control population from the same immune condition. Thus,

$$\bar{d}^{\text{prot}}_{k,c} = \text{mean}\left(X^{\text{prot}}_{k,c}\right) - \mu^{\text{prot}}_{0,c}, \tag{17}$$

and

$$D^{\text{prot}}_{k,c} = \left\{x^{\text{prot}}_{i,k,c} - \mu^{\text{prot}}_{0,c}\right\}_{i=1}^{n_{k,c}}. \tag{18}$$

Fixed RNA-derived rankings. Direction-only mean cosine, magnitude-aware mean L_2 and energy-distance RNA rankings were computed and evaluated against each protein criterion. The primary comparison uses mean L_2 and energy distance, with cosine as a directional reference. RNA populations within an immune condition share a matched control, whose subtraction leaves energy distances unchanged. Queries, candidate libraries, sampled cells and RNA scores were fixed across protein evaluators; protein measurements were used only for evaluation.

Response-magnitude control. The scalar response-magnitude control was

$$M(q, d) = -\left|\left\|\bar{d}^{\text{RNA}}_{q,c}\right\| - \left\|\bar{d}^{\text{RNA}}_{d,c}\right\|\right|. \tag{19}$$

The same magnitude-matching ranking was evaluated against all three protein criteria. Agreement was measured by Spearman correlation across candidate knockouts within each query and summarised by the median across all 659 queries, as well as separately within Control, IFNγ and Co-culture. Paired energy-minus-mean-L_2 differences were calculated within each query before taking their median.

Natural heterogeneity and large-scale validation

Patient-derived glioblastoma and Tahoe-100M were used to measure state-specific responses, their recoverability and agreement of drug ordering across naturally occurring cellular states. The harmonised ZhaoSims2021 dataset[7] contains ten glioblastoma specimen identifiers from acute-slice cultures and surgical biopsies, and six drug treatments: Ana-12, Ispenisib, RO4929097, Tazemetostat, etoposide and panobinostat. Expression counts were normalised to 10^4 per cell, log1p-transformed and reduced to 2,000 highly variable genes. Only one patient received more than two compounds.

Cellular-compartment assignment. Cells were assigned to malignant and myeloid compartments using canonical marker scores; ambiguous cells were excluded. Drug responses were defined relative to patient- and compartment-matched controls. Assignment criteria, retained-cell counts and threshold-sensitivity analyses are reported in Supplementary Note 3.

Differential response in natural tissue. State-specific perturbation responses were measured between malignant and myeloid compartments. For patient p, drug d and compartment k, the response vector was

$$r_{p,d,k} = \text{mean}\left(X^{\text{treated}}_{p,d,k}\right) - \text{mean}\left(X^{\text{control}}_{p,k}\right). \tag{20}$$

Only patient–drug combinations with at least 50 treated cells in both compartments were included. Controls were matched within both patient and compartment. Differential response was summarised as

$$R_{p,d} = \cos\left(r_{p,d,\text{malignant}}, r_{p,d,\text{myeloid}}\right). \tag{21}$$

This construction removes pre-existing differences between patients and cellular compartments before comparing the directions of their drug-induced responses.

Recoverability in natural tissue. Response-identity recoverability was assessed from within-patient drug-pair splits by comparing k-means, Gaussian-mixture, Leiden[28] and HDBSCAN[29] partitions with supervised classification. Patient-level bootstrap and method-pair sensitivity analyses are reported in Supplementary Note 3.

Proxy for downstream decision relevance. We quantified whether drug–drug response similarity is preserved across two disjoint cellular compartments. For drug pair (a, b) measured in the same patient, myeloid-response similarity was defined as

$$S^{\text{myeloid}}_{a,b} = \cos\left(r_{a,\text{myeloid}}, r_{b,\text{myeloid}}\right), \tag{22}$$

and malignant-response similarity as

$$S^{\text{malignant}}_{a,b} = \cos\left(r_{a,\text{malignant}}, r_{b,\text{malignant}}\right). \tag{23}$$

We then computed

$$\rho_{\text{disjoint}} = \rho_S \left(S_{a,b}^{\text{myeloid}}, S_{a,b}^{\text{malignant}} \right), \tag{24}$$

using only drug pairs available within the same patient. The two sides contain no shared cells and each compartment is referenced to its own matched control. This statistic measures agreement between the two compartment-specific drug-pair similarity structures.

Large-scale validation in Tahoe-100M. Tahoe-100M plate 3 was used to test the same requirements at larger scale. We evaluated two independently defined states, released cell-cycle labels and a control-derived k-means partition, and quantified state-specific response alignment, response-identity recoverability and agreement of drug ordering across disjoint states. Dataset filtering and all state-definition details are reported in Supplementary Note 3. For each eligible drug–cell-line condition, the response of state k was defined relative to its own state-matched controls,

$$r_{d,c,k} = \text{mean} \left(X_{d,c,k}^{\text{treated}} \right) - \text{mean} \left(X_{c,k}^{\text{control}} \right). \tag{25}$$

Differential response was measured as the cosine between the two state-specific response vectors,

$$R_{d,c} = \cos \left(r_{d,c,0}, r_{d,c,1} \right). \tag{26}$$

A condition was included only when at least 50 cells were available in each required treated and control state.

Finally, a disjoint state-ordering statistic was used as a large-scale proxy for downstream decision relevance. Within each cell line, cosine similarity was computed for every eligible pair of drug responses separately in the two non-overlapping cellular states. Spearman correlation between the two resulting drug–drug similarity matrices was then calculated,

$$\rho_c = \rho_S \left(S_{a,b}^{(0)}, S_{a,b}^{(1)} \right). \tag{27}$$

For direct candidate ordering, each jointly eligible drug was used as a query against every other jointly eligible drug in the same context. Candidates were ranked by state-specific control-referenced response cosine. First-ranked agreement was defined by overlap of the highest-score candidate sets, including ties; top-five overlap was the shared fraction among the first $\min(5, N)$ candidates, using drug-name order to break ties. Queries with fewer than two candidates were excluded from ordering summaries. Agreement rates were averaged within each context and then summarised by their median across contexts. These comparisons identify closest response analogues. State definitions, cell-count filters and recoverability methods are in Supplementary Note 3.

Source-to-target intervention identification on Tahoe-100M

Task definition and eligibility. The Tahoe-100M plate 3 task identifies the compound that generated a desired treated population from a starting vehicle population. Candidate responses are either observed in the target context or predicted from other contexts. The outcome is the generating compound's rank. A context is one cell line. For a held-out context c and a query drug q, the retriever receives the vehicle (DMSO) cells of c and an observed treated population $X_{q,c}$, and ranks all 92 candidate compounds; the query drug's identity q is hidden from the retriever. A condition entered the grid if it had at least 100 treated and 200 vehicle cells, which admitted 3,992 eligible queries over 44 usable cell lines from the

4,600 conditions present on the plate. Each query was evaluated at five fixed seeds (13, 29, 47, 71, 101), giving 19,960 query–seed pairs. Source and target populations were drawn without replacement at up to 200 cells each, all available cells being taken where a condition held fewer, and responses were referred to the source mean, $P = \{x_i - \bar{x}_{\text{source}}\}$.

Oracle arm. In the oracle arm every candidate's response in the held-out context is taken from the observed data, isolating the retrieval step and defining an information ceiling for prediction. The target population is sampled from one sublibrary half and all candidates, including the generating drug and its 91 competitors, from the other. Each candidate contains up to 100 cells sampled without replacement, using all cells in that half when fewer are available.

Prediction arm. The observed treated population in the held-out context defines the target query. Forward predictors use treated and vehicle cells from the other 43 cell lines, with the held-out context's vehicle population defining the source state for candidate-response prediction. Six predictors were evaluated. Three are additive: they emit one signature per drug and apply it identically to every source cell, so the candidate population is the vehicle population rigidly translated. These are an average effect δ_d pooled over training contexts; a nearest-context transfer taking δ_d from the training context whose vehicle mean is closest in cosine; and a linear-latent variant truncating the average effect to the principal components carrying 80% of training variance. Three allow cell-dependent responses. The *state-conditioned* predictor estimates the same average effect separately for each of the three cell-cycle states released with the plate, G1, S and G2M, and applies it to each source cell according to its state. Each per-state effect is averaged over the training contexts in which that state holds at least 20 cells (359 of the 12,275 cell-line, drug and state groups fall below this floor). Every drug had a usable estimate in every state for every held-out context, so the predictor is non-additive over its whole output. The *Gaussian optimal-transport* map is a closed-form Bures map in a 50-component training principal-component basis. For source and treated latent covariances S_0 and S_1, its linear map is

$$A = S_0^{-1/2} \left(S_0^{1/2} S_1 S_0^{1/2} \right)^{1/2} S_0^{-1/2}. \tag{28}$$

For a held-out source cell x, with P the loading matrix and $z = P^\top (x - \bar{x}_{\text{source}})$, the prediction is $\hat{x} = x + P\{(A - I)z + b_d\}$, where b_d is the average training-context treated-minus-vehicle latent mean shift for drug d. Vehicle and treated clouds are centred on their own condition means before pooling, so S_0 and S_1 estimate within-condition scatter. Each covariance is regularised by adding 10^{-3} times its mean diagonal element to the diagonal. Symmetric matrix square roots use eigendecomposition with eigenvalues floored at 10^{-10}. Each held-out-context basis is fitted on up to 50,000 training cells sampled without replacement, and each context–drug fit uses up to 200 cells. Sampling uses NumPy generators initialised by $[5, 1, c, 0]$ for the held-out-context PCA pool and $[5, 0, 0, 0]$ for condition-level fit cells, where c is the index in the sorted context list. This Gaussian/Bures predictor is distinct from the entropic Sinkhorn optimal-transport diagnostic[30] in Supplementary Note 4. Additional SciPlex3 predictor comparisons, including scGen fitted using scvi-tools[31] and entropic optimal transport computed with POT[32], are reported in Supplementary Note 4 and Supplementary Table 4. *Nearest context, cells* transfers the neighbouring context's actual treated cells for the drug, shifted by the difference of the two vehicle means. This transfers the neighbouring context's observed response distribution.

Scoring and evaluation. Candidates were scored by the three rules of Fig. 1a: a direction-only mean cosine, a magnitude-aware mean L_2, and the population route, primarily the unbiased U-statistic energy distance, with MMD and sliced Wasserstein as secondary variants. The secondary scores use settings specific to this task: sliced Wasserstein over 200 random projections at 100 quantiles, and a single RBF bandwidth fitted once by the median heuristic on the candidate library and held fixed across queries. Mean reciprocal rank of the true intervention is the primary outcome; Hit@1, Hit@5 and Hit@10 are secondary. Intervals are 2,000-fold bootstrap intervals resampled over cell-line clusters, which are the independent units of this design. A decision is counted as corrected when the population route places the true drug first and the reference does not, and as broken in the reverse case.

Headroom, response detectability and the conditional bottleneck analysis

Headroom and its null. Headroom is defined per query as $H = 1 - \mathrm{RR}_{\mathrm{ref}}$, where the reference is direction-only mean-cosine retrieval, and the gain is $G = \mathrm{RR}_{\mathrm{pop}} - \mathrm{RR}_{\mathrm{ref}}$ on the same query. A Spearman correlation between H and G includes the shared reference rank and the reciprocal-rank ceiling. Within each cell line, the population route's reciprocal ranks are randomly permuted across queries, preserving both marginal distributions, the RR ≤ 1 ceiling and the cell-line clustering; G is then recomputed against the unpermuted reference and the correlation retaken. Two hundred permutations at seed 0 give the interval reported in Fig. 5e.

Response detectability. For each held-out cell line c and query drug q, equal numbers of cells treated with q and vehicle cells of c form a balanced two-class problem, at 300 per arm or at the size of the smaller pool where that is below 300. Sampling uses a NumPy generator initialised by $[0, c]$, where c is the sorted context index; logistic cross-validation and clustering use seed 0. Chance accuracy is 0.5. The supervised ceiling A_{sup} is the out-of-fold accuracy of a five-fold cross-validated logistic probe ($C = 1$) on those cells. The unsupervised value A_{unsup} is k-means with $k = 2$ on the same cells, scored by best-permutation accuracy. The labels denote treatment identity defined by the source well. Supervised accuracy provides an empirical reference for response detectability.

Conditional bottleneck regression. The model in Fig. 5h uses population-retrieval reciprocal rank as the outcome and reference mean-cosine reciprocal rank as a covariate. The explanatory variables are interaction ratio, interaction reproducibility $R_{\mathrm{int}} = \cos(I_A, I_B)$ and response detectability A_{sup}; a separate model uses the naive response cosine $1 - \cos(r_1, r_2)$. Conditioning on reference rank accounts for the shared reference in the gain statistic. Estimation used ordinary least squares with cell-line-clustered standard errors. Predictors were standardised to zero mean and unit variance, and complete-case analysis retained 3,618 queries.

Cross-fitted drug-by-state interaction

For the released G1 and G2M states ($s = 1, 2$), let r_s be the treated-minus-vehicle mean response in state s. State-dependent response is measured by $I = r_1 - r_2$. Cross-fitting separates reproducible interaction from sampling noise, which inflates both $\|I\|$ and $1 - \cos(r_1, r_2)$ for weak responses. Each of the four arms is therefore split into disjoint halves A and B, the interaction is estimated once in each, and the two estimates are multiplied:

$$S_{\mathrm{int}} = \frac{\langle I_A,\, I_B \rangle}{p}, \qquad S_{\mathrm{main}} = \frac{\langle M_A,\, M_B \rangle}{p}, \qquad M = \tfrac{1}{2}(r_1 + r_2), \tag{29}$$

with p the number of genes. Independent halves give $E[S_{\text{int}}] = 0$ under the additive null $E[I] = 0$. Both treated and vehicle arms are split to keep control noise independent between halves. Splits use disjoint library preparations when at least two sublibraries are available, and random cell halves otherwise. Arms were capped at 400 cells and required at least 50, and ten A/B splits were averaged per query.

Two derived quantities are used. The *interaction ratio*, $S_{\text{int}}/(S_{\text{int}} + S_{\text{main}})$, uses the cross-fitted numerator and denominator and is defined when $S_{\text{int}} + S_{\text{main}} > 0$; it is unbounded and can be negative or exceed one. The denominator is non-positive in 2 of the 3,620 scored Tahoe conditions, which leaves the 3,618 complete cases in Fig. 5h. $R_{\text{int}} = \cos(I_A, I_B)$ asks whether the interaction reproduces across halves. As a positive control, synthetic conditions were built from the plate's own cells by taking the G1 cells from one drug and the G2M cells from another, so interaction is present by construction in material with the same states and arm sizes as the queries. Across 176 such conditions, the median S_{int} was 7.2 times that of the 3,620 query conditions. The naive cosine $1 - \cos(r_1, r_2)$ is a descriptive comparator in Fig. 5g and the sensitivity regression.

Statistical analysis

Statistical units were defined by the experimental design: cells estimate a perturbation-response population, and retrieval performance is summarised at the designated cluster or task level[33]. The partial-observation arm uses the query drug as the primary inferential cluster. Of the 765 queries, 720 hold out one of 143 drugs and the remaining 45 pool the unmeasured drugs of the partial-library split into a single batch query; those pooled queries remain in descriptive summaries but are excluded from formal drug-cluster inference. The 600 leave-drug-out queries on which mechanism recovery is scored come from 130 drugs. Each drug is scored across three shared cell lines, several library fractions and up to five random seeds, giving 417 distinct drug-by-cell-line contexts in the individual-drug analysis. Primary 95% intervals for this arm use 10,000 percentile bootstrap resamples of query-drug clusters, retaining all rows belonging to each sampled drug. For paired comparisons, repeated observations are first collapsed to one median effect per drug and a two-sided drug-level Wilcoxon signed-rank test is used when reported. For the selected-versus-unselected comparison, inference is summarised by the cluster-bootstrap interval because drugs can contribute to both groups.

Monotonic associations were summarised by Spearman rank correlation. Reported comparisons include effect sizes and the applicable cluster-bootstrap intervals. For the GDSC2 functional analysis, Spearman correlations were computed per query across its candidate drugs. Overall values pool 103 queries descriptively; within-cell-line summaries retain the three cell lines as higher-level units. The protein evaluator-shape experiment was summarised within the Control, IFNγ and Co-culture conditions, its three higher-level units. For the patient-derived glioblastoma analysis, uncertainty in the recoverability gap was estimated by cluster bootstrap at the patient level with 5,000 resamples, differences being taken within each drug-pair split before aggregation; four patients contributed evaluable splits. Intervention-retrieval intervals are 2,000-fold bootstrap resamples over cell-line clusters. Tahoe-100M summaries are taken across eligible drug–cell-line conditions or within-cell-line drug pairs according to the statistic, and both cellular-state definitions, released cell-cycle state and the control-derived k-means partition, were analysed in parallel.

Analyses were implemented in Python. Single-cell data handling used Scanpy[20]; the k-means, Gaussian-mixture, principal-component and logistic-regression procedures named above are the scikit-learn implementations[34]. The GDSC2 mean-vector comparison and the full candidate-ordering analysis

used Python 3.11.14, NumPy 2.2.6, pandas 2.3.3 and SciPy 1.14.1; GDSC2 energy scoring used PyTorch 2.10.0+cu126, and Tahoe state construction used anndata 0.12.10 and scikit-learn 1.7.2.

Data availability

The original source datasets are publicly available as described below. SciPlex3, Frangieh Perturb-CITE-seq and ZhaoSims2021 were obtained in harmonised form from scPerturb[35], Zenodo record 13350497 (doi:10.5281/zenodo.13350497, release 1.4). The exact harmonised files used in the analyses were `SrivatsanTrapnell2020_sciplex3.h5ad` for SciPlex3, `FrangiehIzar2021_RNA.h5ad` and `FrangiehIzar2021_protein.h5ad` for Frangieh Perturb-CITE-seq, and `ZhaoSims2021.h5ad` for the patient-derived glioblastoma analysis. The primary SciPlex3 data are available from the Gene Expression Omnibus under accession GSE139944[6]; the original Frangieh and ZhaoSims2021 datasets are described in the corresponding publications[7,15].

Tahoe-100M[18] was released by Vevo Therapeutics and the Arc Institute under CC0 and is distributed through the Arc Virtual Cell Atlas and Hugging Face (`tahoebio/Tahoe-100M`). This study uses plate 3 only, together with the released `obs_metadata` and `sample_metadata` tables. Analyses use the released preprocessed expression matrix comprising 4,158,278 cells and 2,304 highly variable genes in log1p-transformed space.

The external functional evaluation used GDSC2 fitted dose–response data from the Genomics of Drug Sensitivity in Cancer, release 8.5 (`GDSC2_fitted_dose_response_27Oct23.xlsx`), available from https://www.cancerrxgene.org. The procedure used to match the SciPlex3 and GDSC2 drug panels is described in Methods.

The three core processed matrices and associated annotation tables are publicly available at https://huggingface.co/datasets/Boom5426/PopRetrieve. Selected result tables are available in the code repository.

Code availability

The PopRetrieve software and released experiment drivers are publicly available at https://github.com/Boom5426/PopRetrieve.

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Author contributions

H.W. performed the computational analyses, implemented the methods, curated the data, conducted validation and prepared the visualizations. H.W. wrote the original draft. X.L. and C.C. contributed to data curation, validation and manuscript revision. Q.S. contributed to supervision and manuscript revision. B.L. contributed to study conceptualization, methodology, supervision and manuscript revision. B.Z. supervised the study and contributed to manuscript revision. All authors discussed the results and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary Information is available for this paper: Supplementary Methods, Supplementary Notes 1–4 and Supplementary Tables 1–6.

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